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Experiment:- Study of LAN Topology, Network devices and Address Configuration.

Objective:- →

- 1) To study the LAN topology used in college campus.
- 2) To identify networking device & understand their function.
- 3) To analyze wired & wireless communication method used in the network.
- 4) To measure speed of Network for wired & wireless connections.
- 5) To obtain MAC, IP Address & subnet mask.

Problem Statement -1

• Draw & identify LAN Topology of college campus & clearly label all components such as routers, switches & etc.

→ Solⁿ → The college campus network observed this study consists of:

- * Layer-3 Core switches in the server Room.
- * Managed by PoE switches.
- * Wireless Access point (APs)
- * Ethernet port in classroom & laboratories.

Network Topology

Hybrid Hierarchical Topology which combines.

- 1) Tree Topology
- 2) Extended star Topology.
- 3) Ring Topology (Server room).

②

The network hierarchical layer

- 1) Core layer.
- 2) Distribution layer.
- 3) Access layer

Core Layer

Core layer contains Server Room and Server room have Firewall, Router, 2-3+ switches, Patch Rack, Server, inverter, etc.

Distribution Layer

* Distribution layer contains 4-switches.

* The 4 switches are →

- ① Main Build → R7 (Room no-213, 2nd floor)
- ② Main Build → R1 (Room no-301, 3rd floor).
- ③ Main Build → R6 (Room no-113, 1st floor)
- ④ Main Build → R2 (Room no-201, 2nd floor).

* This 4-switches contains →

- ① 18 Port Gigabit Easy smart switch with 16-port.

TL-SG1218MPE

- ② SG13428XMP → 24 Port Gigabit & 4-port 10 GbESFP+, L2+ managed switches with 24 Port PoE+.

→ Now let's discuss these 2 above components →

- ① TP-Link TL-SG1218MPE

* Likely used for: CCTV camera, wifi access point, etc.

* 18 Port Gigabit "Easy Smart" switch.

* 16 Port Support PoE+

② TP-Link Omada S6342 8xMP

- * Higher-end managed Layer 2 switch.

- * 24 PoE+ port, Uplinks, 10G SFP

- * Usually act as a main distribution for each floor.

→ Access Layer

- * Access Layer contain 3-switches i.e

- 1) Main Build R1 - [Room no-301, 3rd floor],

- 2) Main Build R4 [Room no-05, Ground floor],

- 3) Main Build R5 [Room no-18, Ground floor].

- * Every Switches act as access layer which helps to provide network connectivity to all Classroom, Lab, Faculty Room, etc.

- * Access Layer directly Connect:

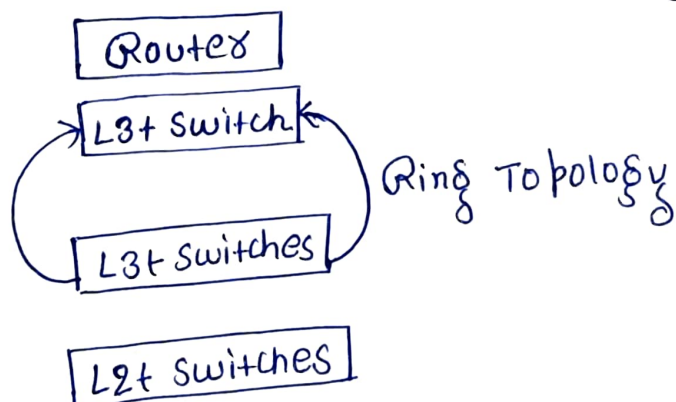
- ① Ethernet Port

- ② Wireless AP

- ③ Lab system

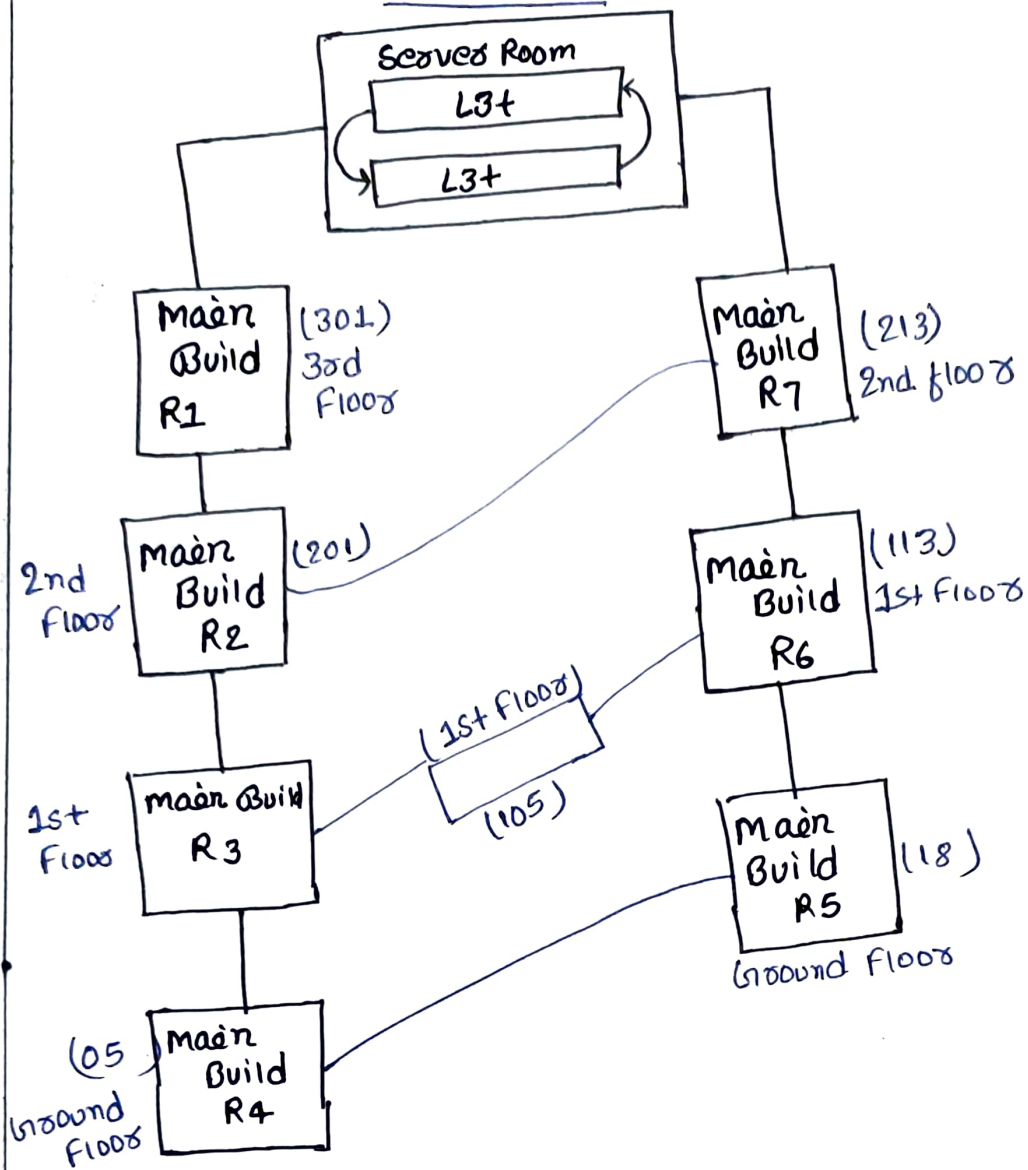
- ④ Outdoor wireless AP (ex-D18)

Server Room Architecture / Topology .



(A)

TOPOLOGY

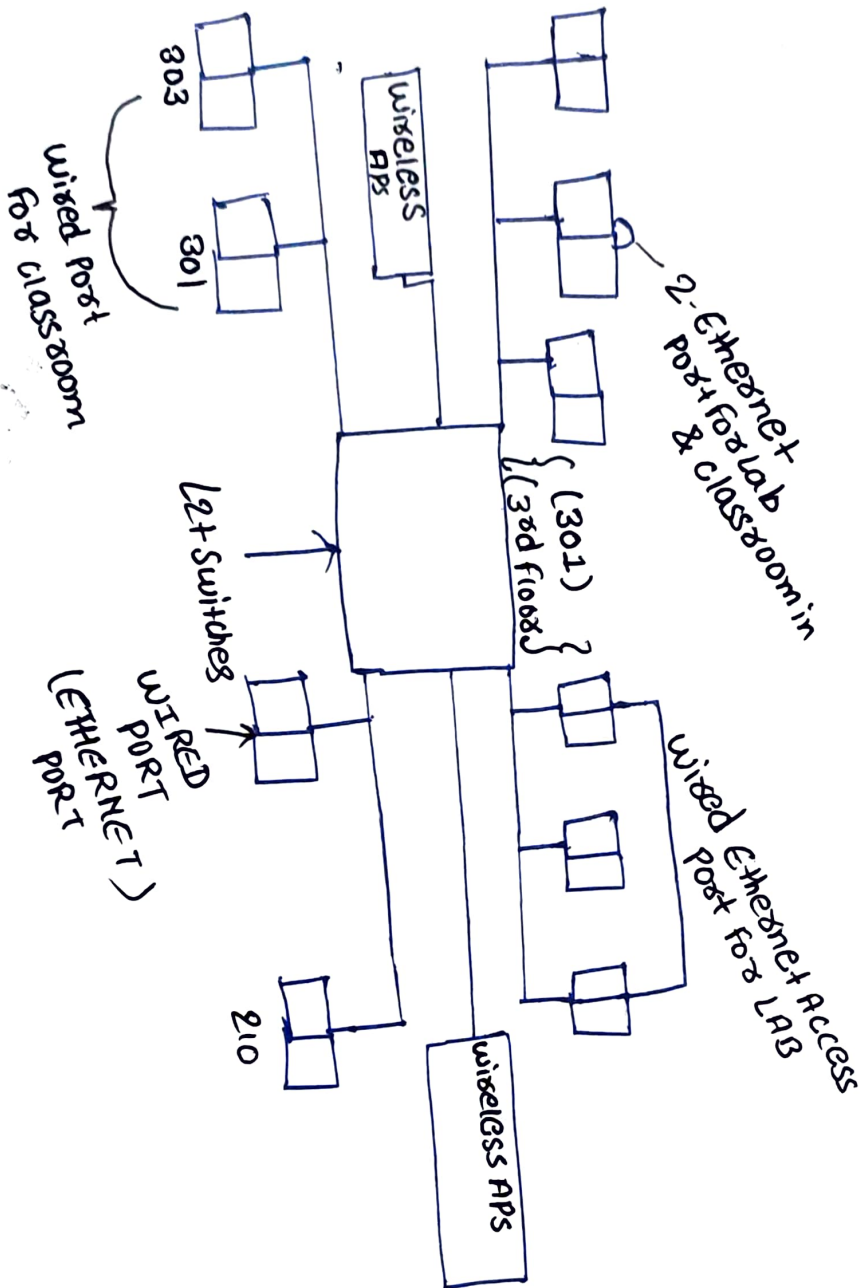


→ R1, R7, R2, R6 → Distribution layer

→ Server room → Core layer

→ R1, R2, R3, R4, R5, R6, R7 → Access layer

ACCESS LAYER SWITCH (3rd Floor - 301)



Windows PowerShell

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Install the latest PowerShell for new features and improvements! <https://aka.ms/PSWindows>

PS C:\Users\LENOVO> netsh wlan

The following commands are available:

Commands in this context:

?	- Displays a list of commands.
add	- Adds a configuration entry to a table.
connect	- Connects to a wireless network.
delete	- Deletes a configuration entry from a table.
disconnect	- Disconnects from a wireless network.
dump	- Displays a configuration script.
export	- Saves WLAN profiles to XML files.
help	- Displays a list of commands.
IHV	- Commands for IHV logging.
refresh	- Refresh hosted network settings.
reportissues	- Generate WLAN smart trace report.
set	- Sets configuration information.
show	- Displays information.
start	- Start hosted network.
stop	- Stop hosted network.

To view help for a command, type the command, followed by a space, and then type ?.

PS C:\Users\LENOVO> netsh wlan show interface

There is 1 interface on the system:

```
PS C:\Users\LENOVO> netsh wlan show interface
```

There is 1 interface on the system:

Name	: Wi-Fi
Description	: Intel(R) Wi-Fi 6 AX203
GUID	: b0dd9445-2ec8-4691-a2f9-f04a0b8ecad9
Physical address	: ae:39:d7:98:56:7d
Interface type	: Primary
State	: connected
SSID	: GECK-Campus-WiFi
AP BSSID	: b0:19:21:74:07:b9
Band	: 5 GHz
Channel	: 48
Connected Akm-cipher	: [akm = 00-0f-ac:02, cipher = 00-0f-ac:04]
Network type	: Infrastructure
Radio type	: 802.11ac
Authentication	: WPA2-Personal
Cipher	: CCMP
Connection mode	: Auto Connect
Receive rate (Mbps)	: 390
Transmit rate (Mbps)	: 433.3
Signal	: 85%
Rssi	: -69
Profile	: GECK-Campus-WiFi
QoS MSCS Configured	: 0
QoS Map Configured	: 0
QoS Map Allowed by Policy	: 0

```
PS C:\Users\LENOVO> ipconfig
```

Windows IP Configuration

```
PS C:\Users\LENOVO> ipconfig
```

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Link-local IPv6 Address : fe80::db30:b3c7:411a:92e1%13
IPv4 Address. : 10.85.18.117
Subnet Mask : 255.255.248.0
Default Gateway : 10.85.16.1

```
PS C:\Users\LENOVO> getmac
```

Physical Address	Transport Name
AE-39-D7-98-56-7D	\Device\Tcpip_{B0DD9445-2EC8-4691-A2F9-F04A0B8ECAD9}

```
PS C:\Users\LENOVO>
```

Wi-Fi

Intel(R) Wi-Fi 6 AX203

Throughput

5 Mbps

3.6 Mbps

60 seconds

0

Send

8.0 Kbps

Receive

256 Kbps

Adapter name: Wi-Fi

SSID: K

Connection type: 802.11n

IPv4 address: 172.31.74.145

IPv6 address: fe80::d305:5f2d:96c9:fd76%13

Signal strength: 

⑥ Difference b/w wired & wireless speed?

Soln

Wired Network	Wireless Network
* Faster Communication	* Slower Communication
* Stable Connection	* Affected by distance / Interference
* Low latency	* Higher latency
* Dedicated Physical medium	* Shared radio medium.

→ Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.

→ Wireless connections use radio waves which are affected by interference, distance, obstacles & multiple connected users.

Q) Role of Subnet mask in network?

→ Soln :-

A subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

- ① Network Segment
- ② Efficient routing
- ③ Communication b/w devices.

⑦

OBSERVATION,

During the study of the college network infrastructure:-

- ① A centralized server room was used of network backbone.
- ② Two L3 switches were connected in redundant cyclic.
- ③ Managed TP-Link PoE+ switches were used in network cabinets.

Advantage of Existing Network?

- * Scalable architecture.
- * Centralized management.
- * Supports PoE+ device
- * Easy expansion of floors & classroom.
- * Support both wireless & wired.
- * Redundant link for failover.

Conclusion :-

The observed campus LAN follow a hybrid hierarchical topology consist of Tree, Extended star, Ring. The network uses managed PoE switches, wireless access point, Ethernet port & centralized server infrastructure to provide efficient wired & wireless communication across the campus.